torchgeo-bench

Two-Week Sprint Recap

~50 PRs · GPU evaluation · OlmoEarth v1.1 · calibration · efficiency profiling · leaderboards

May 2026

What shipped

GPU GPU KNN via faissknn

FAISS-backed KNN with optional CUDA path. Auto-fallback to CPU.

CLI Typer + Rich

Replaced argparse + tqdm. Beautiful progress bars, rich tracebacks.

PROFILE Efficiency Profiling

Throughput, GFLOPs, peak GPU mem, energy (Wh/1k), $/inference.

MODELS OlmoEarth v1 + v1.1

nano→large, plus v1.1 linear-embed family. DINOv3-SAT ViT-L web-pretrained.

CALIB Calibration Metrics

ECE · RMS-CE · MCE + temperature scaling on every probe.

DATA EuroSAT Spatial Split

Geographically disjoint train/test. Harder than random split.

QUALITY Cleanlab Label Audit

Label-quality scores across all GeoBench V1+V2 datasets.

POOL CLS + Mean Pool Ablations

TerraTorch/ScaleMAE/Clay — pool=cls|mean|both sweep.

FIX Silent-bug Sweep

Removed try/except covers; fixed minmax_zscore, fp16 overflow, label gaps.

MULTI SAR + Landsat Modalities

OlmoEarth mixed-sensor support, auto input-resolution.

GPU KNN — faissknn

PR #53 PR #55 PR #89 PR #94 PR #101

CPU path:

```
clf = KNNClassifier(n_neighbors=5, device="cpu")
clf.fit(x_train, y_train)
preds = clf.predict(x_test)
```

GPU path (opt-in via `pip install -e ".[cuda]"`):

```
clf = KNNClassifier(
    n_neighbors=5,
    device="cuda:0",
    metric="cosine",  # l2 | ip | cosine
)
clf.fit(x_train, y_train)
preds  = clf.predict(x_test)
proba  = clf.predict_proba(x_test)
```

Key fixes:

- `n_classes = max(y)+1` not `len(unique(y))` — avoids `IndexError` on partitions with missing class labels
- `use_fp16=False` in evaluation — raw sensor DN values (~10 000) overflow fp16 L2 distances → random KNN
- `faiss-cuda-cu128` now the **sole** core backend — `manylinux_2_28` wheels run on CPU *and* GPU, killing the old faiss-cpu namespace clash (`#101`)

Config knobs (`#94` , `#99`):

```
eval:
  knn_k: 5          # neighbours
  knn_device: null  # null → inherit cfg.device
                    # "cpu" forces faiss-cpu KNN
```

Typer + Rich CLI

PR #88

Before (argparse + tqdm):

```
parser = argparse.ArgumentParser()
parser.add_argument("--model", ...)
args = parser.parse_args()

for batch in tqdm(dataloader):
    ...
```

After (Typer + Rich):

```
app = typer.Typer(no_args_is_help=True)

@app.command(context_settings={
    "allow_extra_args": True,
    "ignore_unknown_options": True,
})
def run(ctx: typer.Context) -> None:
    """Run benchmarks; extra args → Hydra."""
    sys.argv = [sys.argv[0], *ctx.args]
    hydra_main()
```

Rich progress bars:

```
from rich.progress import track

for batch in track(dataloader,
                   description="Extracting"):
    features = model(batch["image"])
```

Rich logging:

```
from rich.logging import RichHandler

logging.basicConfig(handlers=[
    RichHandler(rich_tracebacks=True)
])
```

Rich tables (tune_dataloader.py):

```
from rich.table import Table
table = Table(header_style="bold cyan")
```

Efficiency Profiling

PR #60 PR #62 PR #63 PR #67

Metrics recorded per model run:

```
{
  # GPU
  "throughput_samples_per_sec": 1420.3,
  "peak_gpu_mem_gb":            3.2,
  "gpu_power_w_avg":            182.0,
  "energy_wh_per_1k_samples":   0.036,
  "gflops":                     61.6,
  "params_m":                   307.4,
  # CPU
  "throughput_samples_per_sec_cpu": 42.1,
  # Cost
  "cost_usd_per_1M_samples":    0.12,
  "gco2_per_1M_samples":        18.4,
}
```

GFLOPs via FlopCounterMode:

```python
with torch.profiler.FlopCounterMode() as fc:
    model(sample)
gflops = fc.get_total_flops() / 1e9
```

Energy via pynvml:

```python
import pynvml
pynvml.nvmlInit()
h = pynvml.nvmlDeviceGetHandleByIndex(0)
mw  = pynvml.nvmlDeviceGetPowerUsage(h)
wh  = (mw / 1000) * (elapsed_s / 3600)
```

Cost extrapolation:

```python
# A100 spot ~$1.50/hr on Lambda
cost = (1_000_000 / throughput) / 3600 * 1.50
```

OlmoEarth Integration

PR #84 PR #85 PR #93

Config (nano → large):

```yaml
# conf/model/olmoearth_v1_base.yaml
_target_: torchgeo_bench.models.OlmoEarthBench
name: olmoearth_v1_base
variant: base
normalization: identity  # model handles its own
```

Auto-rescale to S2 DN:

```python
class OlmoEarthBench(BenchModel):
    expected_input_unit = InputUnit.S2_DN

    def _forward_patch_features(
        self, images: Tensor, **_
    ) -> Tensor:
        # model_native normalizer already
        # rescaled inputs to S2 DN range
        return self.backbone(images)
```

m-eurosat Linear Accuracy:

	Model	Score
1	OlmoEarth Large	**0.976**
2	OlmoEarth Base	**0.975**
3	DOFA Large	0.973
4	OlmoEarth v1.1 Base	0.970
5	DINOv3-SAT ViT-L	0.969

Dominates S2 datasets — EuroSAT, BigEarthNet, So2Sat. Nano/Tiny competitive at a fraction of the size.

OlmoEarth v1.1

PR #99

What changed (v1 → v1.1):

- **Linear** patch embedding (vs. convolutional)
- Single bandset per modality
- Updated masking + loss functions

≈ 3× FEWER MACS at comparable accuracy.

```
# conf/model/olmoearth_v1_1_base.yaml
name: olmoearth_v1_1_base
version: v1.1      # selects weight family
```

30/30 sweep tasks · 0 failures · Nano/Tiny/Base.

Linear probe — v1 vs v1.1

Dataset	Metric	v1	v1.1
m-eurosat (Base)	Acc	.975	.970
m-so2sat (Base)	Acc	.720	**.728**
m-so2sat (Tiny)	Acc	.656	**.693**
m-bigearthnet (Tiny)	mAP	.691	**.717**
benv2 (Tiny)	mAP	.817	**.826**
treesatai (Base)	mAP	.647	.645

Smaller variants gain most — Tiny/Nano jump on So2Sat + BigEarthNet (KNN So2Sat Tiny .506 → **.567**). Slight EuroSAT dip. Net: same accuracy at a third of the compute.

CLS Token + Pool Ablations

PR #73 PR #74 PR #75 PR #76

`pool` kwarg across TerraTorch wrappers:

```
class TerramindBench(BenchModel):
    def _forward_patch_features(
        self, images: Tensor, **_
    ) → Tensor:
        out = self.model.encode(images)
        if self.pool == "cls":
            return out[:, 0]
        elif self.pool == "mean":
            return out[:, 1:].mean(1)
        else:  # "both"
            return torch.cat([
                out[:, 0],
                out[:, 1:].mean(1),
            ], dim=1)
```

Config variants:

```
name: tt_clay_v1_5_base_cls   # pool: cls
```

Finding: CLS helps on ForestNet / Brick Kiln; patch-mean wins on EuroSAT / BigEarthNet.

Terramind has no CLS token — `_cls` configs dropped (`#76`).

m-brick-kiln Linear Accuracy:

	Model	Score
1	DINOv3-SAT ViT-L	**0.976**
2	Clay v1.5 Base+CLS	**0.975**
3	DOFA Base	0.974
4	DOFA Large	0.974

EuroSAT Spatial Split + Cleanlab

PR #50 PR #52

Geographically disjoint train/test:

```
# Standard: random split
# → models exploit spatial autocorrelation

# Spatial: tiles from different regions
# → true generalization test
ds = EuroSATSpatialBench(
    root=DATA_ROOT,
    partition="default",
    bands=["B02","B03","B04","B08"],
)
```

OlmoEarth Large KNN: **0.959** vs 0.956 random split
OlmoEarth Base Linear: **0.978** vs 0.975 random split

Cleanlab label audit:

```
from cleanlab.filter import find_label_issues

issues = find_label_issues(
    labels=y_train,
    pred_probs=pred_proba,
    return_indices_ranked_by=
        "self_confidence",
)
# → flags likely mislabeled samples
```

Applied across all GeoBench V1+V2. Results in `results/cleanlab/` .

Surfaces annotation noise in BigEarthNet, ForestNet, TreeSatAI — useful for reweighting or curriculum training.

Leaderboard — EuroSAT

RESULTS

m-eurosat (Accuracy)

	Model	KNN	Lin
1	OlmoEarth Large	.956	.976
2	OlmoEarth Base	.946	.975
3	Panopticon	.948	.968
4	DOFA Large	.936	.973
5	OlmoEarth v1.1 Base	.930	.970

eurosat-spatial (Accuracy)

	Model	KNN	Lin
1	OlmoEarth Large	.959	.977
2	OlmoEarth Base	.941	.978
3	Panopticon	.930	.962
4	OlmoEarth Tiny	.928	.963
5	ResNet50-RGB MoCo	.933	.958

Leaderboard — BigEarthNet

RESULTS

m-bigearthnet (micro-mAP)

	Model	KNN	Lin
1	OlmoEarth v1.1 Base	.664	.771
2	OlmoEarth Large	.664	.764
3	OlmoEarth Base	.658	.769
4	Panopticon	.652	.735
5	Terramind Large	.628	.750

benv2 (micro-mAP)

	Model	KNN	Lin
1	OlmoEarth Base	.735	.853
2	OlmoEarth v1.1 Base	.734	.852
3	OlmoEarth Large	.728	.850
4	Terramind Large	.712	.846
5	OlmoEarth v1.1 Tiny	.725	.826

Leaderboard — ForestNet + So2Sat

RESULTS

m-forestnet (Accuracy)

	Model	KNN	Lin
1	DINOv3-SAT ViT-L	.425	.582
2	Panopticon	.427	.550
3	ScaleMAE Large+CLS	.403	.569
4	Clay v1.5 Base+CLS	.414	.556
5	Clay v1.5 Base	.395	.551

Hard dataset — all models below .60

m-so2sat (Accuracy)

	Model	KNN	Lin
1	OlmoEarth v1.1 Base	.606	.728
2	OlmoEarth Base	.577	.720
3	OlmoEarth v1.1 Tiny	.567	.693
4	OlmoEarth Large	.562	.689
5	Panopticon	.532	.693

Linear leader: Terramind Base .739 — biggest KNN/Lin gap in the set

Leaderboard — Brick Kiln + PV4GER + TreeSatAI

RESULTS

m-brick-kiln (Acc)

	Model	KNN	Lin
1	DOFA Large	.969	.974
2	DOFA Base	.965	.974
3	Clay Base	.964	.972
4	Clay Base+CLS	.960	.975
5	OlmoEarth Base	.945	.970

m-pv4ger (Acc)

	Model	KNN	Lin
1	DINOv3 ViT-L	.964	.974
2	DOFA Large	.967	.969
3	DOFA Base	.965	.966
4	ScaleMAE+CLS	.956	.970
5	Panopticon	.958	.968

treesatai (mAP)

	Model	KNN	Lin
1	DINOv3-SAT ViT-L	.477	.682
2	Clay Base+CLS	.475	.671
3	DINOv3 ViT-L	.473	.664
4	DOFA Large	.474	.647
5	OlmoEarth Base	.469	.647

Efficiency — Throughput vs Accuracy

PR #60–#80

GPU throughput (img/s) — m-eurosat

Model	img/s	GFLOPs	Acc
ResNet-50 MoCo	3 193	8	.76
DOFA Base	1 747	36	.71
Terramind Base	1 680	36	.68
OlmoEarth Tiny	780	9	.73
OlmoEarth Nano	789	2	.67
DOFA Large	581	124	.73
DINOv3-SAT ViT-L	346	165	.74

Pareto surprises:

WINNER **ResNet-50 MoCo** — 3 193 img/s, 8 GFLOPs, 23M params, yet **.76 avg acc.** Matches or beats every ViT-scale model while running 5–10× faster.

EFFICIENT **OlmoEarth Nano** — 3.6M params, 0.6 GB peak VRAM, 1.6 GFLOPs. Matches Panopticon accuracy at 5× lower cost.

EFFICIENT **OlmoEarth Tiny** (14M) ties DOFA Large (337M) and DINOv3-SAT (304M) at 2× the throughput.

Worst value: Prithvi v1/v2 — 1 700 img/s, 35 GFLOPs, only .56–.60 acc. Same throughput tier as DOFA Base but far lower accuracy.

Intrinsic Dimensionality Analysis

METHOD: INTRINSIC_DIM

id_TwoNN avg · norm. acc = rank-normalized within each dataset

Model	id_TwoNN	norm. acc	Feat dim
EarthLoc ResNet-50	**44**	27%	4 096
OlmoEarth Large	22	85%	1 024
Prithvi v2 100M CLS	18	53%	768
DOFA Large	17	85%	1 024
DINOv3 ViT-L	17	72%	1 024
DINOv3-SAT ViT-L	16	77%	1 024

Pattern: High intrinsic dim ↔ high norm. accuracy. Models with id > 14 consistently score 72–85% norm. acc. Norm. acc = per-dataset rank-normalized, then averaged — removes scale differences between datasets.

Outliers:

TASK GAP **EarthLoc ResNet-50** — id = 44 (highest), but only .59 acc. Trained for geo-localization, not classification. High dimensionality without discriminative structure.

SURPRISE **OlmoEarth Nano** — 128-d output yet id_TwoNN = 14, matching DINOv3-SAT ViT-L (1024-d). Packs equivalent intrinsic complexity into 8× fewer dimensions.

Calibration Metrics

PR #105

ECE / RMS-CE / MCE on every classification probe, plus a temperature-scaling baseline (Guo et al., 2017):

```yaml
eval:
  calibration:
    n_bins_knn: null   # null → knn_k + 1
    n_bins_linear: 15
    temp_scale: true   # +ece_ts / mce_ts
                       #   + temperature
```

```python
# fit T on val logits, NLL via LBFGS
T = fit_temperature(val_logits, y_val)
# T > 1 overconfident · T < 1 under
```

KNN probs quantize to k+1 levels → binned at knn_k+1. TS applies to Linear only.

Linear probe — m-eurosat (RGB)

Model	ECE	ECE·TS	T
DOFA Large	**.007**	.021	0.23
DINOv3-SAT ViT-L	.016	.030	0.18
ResNet-50 MoCo	.018	.032	0.34
OlmoEarth Base	.032	.054	0.71
DOFA Base	.043	**.031**	0.21

FINDING Probes are already well-calibrated (ECE < .05). Fitted **T < 1** → logits *underconfident*, so temperature scaling often **raises** ECE. DOFA Base is the lone beneficiary.

Key Findings

S2 multispectral (EuroSAT, BigEarthNet, So2Sat)
→ **OlmoEarth** dominates KNN + Linear

RGB / aerial (ForestNet, PV4GER, Brick Kiln)
→ **DINOv3-SAT** or **Panopticon** lead

Multi-label (BigEarthNet, BENv2, TreeSatAI)
→ OlmoEarth top KNN; Terramind competitive linear

⚡ Terramind **Base** beats Terramind **Large** on So2Sat linear (0.739 vs 0.712)

⚡ CLS token **collapses** on EuroSAT for Prithvi/Clay — patch-mean wins

⚡ **ImageStats** (raw pixel stats) still hits .95 on m-brick-kiln — low semantic difficulty, though GeoFMs now edge it out

⚡ **SatlasPretrain Swin** (RGB-only) peaks at m-pv4ger #9 / m-brick-kiln #16 — RGB backbones trail multispectral FMs on S2

⚡ **OlmoEarth v1.1** matches v1 at ≈ **3× fewer MACs**; Tiny/Nano gain most on So2Sat + BigEarthNet

⚡ Linear probes are **well-calibrated out of the box** (ECE < .05) — temperature scaling rarely helps